

January 2026 CSE 220

Online on DFT and FFT

Frequency-Domain Image Steganography

Subsections: C1, C2

Total Marks: 10

Total Time: 40 Minutes

Frequency-Domain Image Steganography

Frequency-domain steganography hides information by modifying selected frequency components of a cover image. Low-frequency components generally represent broad, slowly varying image content, while higher-frequency components capture finer details and edges. In this task, the secret image is embedded into selected high-frequency components of the cover image. You will use your implementation of DFT/FFT to implement a simple steganography pipeline.

Mathematical Model

Let the Fourier spectra of the cover image and secret image be C and M , respectively. Let a be the embedding strength and let K denote the provided high-frequency mask.

The stego spectrum S is constructed as

$$S = (1 - a)C + aM$$

for the selected high-frequency components.

For all other frequency components, the original cover spectrum is retained:

$$S = C.$$

Thus, the complete operation can be expressed as

$$S[K] = (1 - a)C[K] + aM[K]$$

while

$$S[!K] = C[!K].$$

The final stego image is obtained by applying the inverse 2D transform to S and keeping the real-valued component.

The cover image determines the transform size. The secret image must not be larger than the cover image; it is padded to the cover image's dimensions before its transform is computed.

Tasks

1. Complete Fourier transforms.

Compute the Fourier transform of the padded cover image and the padded secret image using implementation of DFT/FFT. **[3 Marks]**

2. Complete frequency-domain embedding.

Use the provided high-frequency mask and embedding equation to modify **only the selected frequency components** of the cover spectrum. The embedding must use the supplied **strength** value and leave all unselected components unchanged. **[4 Marks]**

3. **Complete inverse transform.**

Perform the inverse 2D transform on the modified stego spectrum. Keep the real-valued component to obtain the final stego image. **[3 Marks]**